

Ryu–Takayanagi area law versus a microscopic horizon entropy with $-(3/2) \log(A/4G_N)$ correction: a Lean 4 inconsistency note

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We formalize, in the Lean 4 proof assistant, the structural inconsistency between the classical Ryu–Takayanagi area law [1] $S = A/(4G_N)$ (with no log correction) and the Kaul–Majumdar microscopic horizon entropy [5] $S = A/(4G_N) - (3/2) \log A/(4G_N)$ (with a universal $-3/2 \log$ coefficient). The two formulas agree only on the knife-edge case $A = 4G_N$ (reduced area unity); otherwise they necessarily differ by exactly the log-correction term. We encode the Ryu–Takayanagi formula as an external-hypothesis structural predicate `H_RT_Formula_Valid` (because the bulk holographic dual is not constructed in this project) and the universal 2D-CFT entanglement-entropy log bound $S(L) \leq (c/3) \log(L/\varepsilon)$ of Holzhey–Larsen–Wilczek [2] and Calabrese–Cardy [3]—with Casini–Huerta [4] as the free-QFT specialization—as a similar predicate `H_CasiniHuerta_Bound_Valid` (the predicate name retains the historical `H_CasiniHuerta_*` prefix used in the Lean module; see footnote ??). The substantive result is a knife-edge biconditional locating classical-RT/Kaul–Majumdar agreement at the unique reduced-area value $A/(4G_N) = 1$, plus a numerical anchor pinning the gap at canonical reduced area $A/(4G_N) = 2$ to exactly $(3/2) \log 2 \approx 1.040$. A substantive cross-bridge consumes both the external Ryu–Takayanagi hypothesis and the Kaul–Majumdar entropy function, deriving the inconsistency at any non-trivial reduced area; a falsifier proves that the Kaul–Majumdar function fails the Ryu–Takayanagi hypothesis. The structural conclusion is that classical Ryu–Takayanagi must be augmented with a quantum log correction at the area-log level for consistency with the Kaul–Majumdar microscopic count. The bulk minimal-surface construction [6] and the Casini–Huerta proof from modular-Hamiltonian methods are deferred. The module ships 7 substantive theorems with zero `sorry` statements and zero new axioms.

I. INTRODUCTION

Two distinct frameworks for black-hole and entanglement entropy agree on leading-area behavior but disagree at logarithmic order. Ryu and Takayanagi’s holographic area law [1], $S_{\text{RT}} = A/(4G_N)$, identifies entropy with a bulk minimal-surface area; this is the leading classical statement, with no log correction. Kaul and Majumdar’s microscopic horizon entropy [5],

$$S_{\text{KM}}(A, G_N) = A/(4G_N) - \frac{3}{2} \log A/(4G_N), \quad (1)$$

includes a universal $-3/2 \log$ coefficient extracted from the microscopic counting of horizon Chern–Simons states. A separate non-universality witness in the literature is Sen’s 4D Schwarzschild log coefficient $77/45 \approx 1.711$ [7], which differs from $-3/2$, indicating that the microscopic log coefficient is not universal across Lagrangian regularizations.

This paper presents a Lean 4 formalization of the structural content of this disagreement. We do *not* derive the Ryu–Takayanagi formula from first principles; the bulk holographic dual is not constructed in this project. Instead, we encode it as an external-hypothesis structural predicate and ask: what structural consequences follow when it is combined with a microscopic horizon entropy carrying a log correction?

The output is a small but precise theorem set: a knife-edge biconditional locating the unique reduced area where the two formulas agree, a numerical anchor pinning the gap at canonical reduced area $A/(4G_N) = 2$ to $(3/2) \log 2 \approx 1.040$, a cross-bridge proving inconsis-

tency at every non-trivial reduced area, and a falsifier proving that the Kaul–Majumdar function fails the Ryu–Takayanagi hypothesis directly.

II. EXTERNAL-HYPOTHESIS PREDICATES

a. Ryu–Takayanagi.

```
structure H_RT_Formula_Valid
  (S_BH : R -> R -> R) : Prop where
  rt_proportional :
    forall A G_N, 0 < A -> 0 < G_N ->
      S_BH A G_N = A / (4 * G_N)
```

A candidate entropy function $S_{\text{BH}} : \mathbb{R} \times \mathbb{R} \rightarrow \mathbb{R}$ satisfies `H_RT_Formula_Valid` if and only if it equals $A/(4G_N)$ on the positive cone of (A, G_N) .

b. Casini–Huerta.

```
structure H_CasiniHuerta_Bound_Valid
  (S_ent : R -> R)
  (c_central UV_cutoff : R) : Prop where
  c_pos : 0 < c_central
  uv_pos : 0 < UV_cutoff
  ch_bound :
    forall L, UV_cutoff < L ->
      S_ent L <=
        (c_central / 3) * Real.log (L / UV_cutoff)
```

For a 2D conformal-field-theory entanglement entropy with central charge c and UV cutoff ε , the universal log bound [2, 3] asserts $S_{\text{ent}}(L) \leq (c/3) \log(L/\varepsilon)$ on the substantive case $L > \varepsilon$ [?].

III. STRUCTURAL CONSEQUENCES OF RYU-TAKAYANAGI

Theorem (*rt_entropy_pos*). Under `HRTFormula.Valid`, the entropy is strictly positive on the positive cone of (A, G_N) . The proof rewrites via the hypothesis and applies positivity of division.

Theorem (*rt_kaulMajumdar_gap_at_reduced_area_two*). At canonical reduced area $A/(4G_N) = 2$ (equivalently $A = 8G_N$), the gap between classical Ryu–Takayanagi and the Kaul–Majumdar microscopic entropy is exactly $(3/2)\log 2 \approx 1.040$. The proof unfolds the Kaul–Majumdar definition, normalizes the reduced-area arithmetic, and applies `ring`.

IV. KNIFE-EDGE BICONDITIONAL

The structural anchor of this paper is the knife-edge biconditional

theorem

```
rt_eq_kaulMajumdar_iff_trivial_
  reduced_area
  (A G_N : R) (hA : 0 < A) (hG : 0 < G_N) :
  A / (4 * G_N) = kaulMajumdarS A G_N 0
  <-> A / (4 * G_N) = 1
```

locating classical-Ryu–Takayanagi / Kaul–Majumdar agreement at the unique reduced area $A/(4G_N) = 1$. By contrapositive, the two formulas necessarily disagree at every non-trivial reduced area.

The forward direction proceeds by extracting $\log A/(4G_N) = 0$ from the assumed equality, applying $\exp \circ \log = \text{id}$ on positives to derive $A/(4G_N) = 1$, and discharging. The reverse direction collapses via $\log 1 = 0$.

Theorem (*rt_falsified_by_kaul_majumdar*). Under `HRTFormula.Valid` for a candidate S_{BH} , the Kaul–Majumdar entropy function is structurally incompatible with S_{BH} at non-trivial reduced area. The proof body invokes both the hypothesis (rewriting S_{BH} to $A/(4G_N)$) and the contrapositive of the biconditional (deriving the inconsistency).

V. CASINI-HUERTA BOUND POSITIVITY AND A WITNESS

The Casini–Huerta $\log$ bound’s positivity ships as `ch_log_bound_pos_at_log_pos`: under `H.CasiniHuerta.Bound.Valid`, the bound $(c/3)\log(L/\varepsilon)$ is strictly positive on the substantive case $L > \varepsilon > 0$. The proof combines the structural hypothesis fields $(c > 0, \varepsilon > 0)$ with logarithm positivity and multiplication of positives.

A concrete witness ships as `H.CasiniHuerta.Bound.Valid.witness_saturated`:

the saturated entropy function

$$S_{\text{sat}}(L) := (c/3)\log(L/\varepsilon) \quad (2)$$

satisfies the Casini–Huerta hypothesis with the bound met at equality. This pins the hypothesis as non-vacuous.

VI. FALSIFIER: KAUL-MAJUMDAR FAILS RYU-TAKAYANAGI

The Kaul–Majumdar entropy function fails `HRTFormula.Valid` (theorem `kaulMajumdar_not_HRT`): at $A = 8$ and $G_N = 1$, the microscopic entropy is $2 - (3/2)\log 2 \approx 0.96$, not the $2 = A/(4G_N)$ that Ryu–Takayanagi requires. The proof extracts the discrepancy via $\log 2 > 0$ and a linear-arithmetic step. This is the sharp distinction between microscopic-quantum-corrected entropy and classical-area-law entropy.

VII. FORMALIZATION SUMMARY

The Lean module `lean/SKEFTHawking/RTCasiniHuertaBounds.1` ships 7 substantive theorems, zero `sorry` statements, and zero new axioms beyond the kernel’s three. The Python mirror in `src/rt_ch_bounds/` provides numerical evaluation of the Ryu–Takayanagi–Kaul–Majumdar gap and a 29-case `pytest` suite cross-checking the Lean theorems against numerical agreement.

VIII. WHAT IS AND IS NOT DONE HERE

The formalization deliberately stays on the structural side. It does not:

- Construct the bulk minimal surface; the Lewkowycz–Maldacena [6] replica-trick derivation of Ryu–Takayanagi is not formalized.
- Derive the Casini–Huerta bound from modular Hamiltonian methods; the bound enters as a structural hypothesis.
- Identify which conformal field theory on the boundary corresponds to a given microscopic substrate; the AdS/CFT spectrum dictionary is not constructed.

These are natural extensions but require substantial additional infrastructure (replica-trick formalization, modular-Hamiltonian machinery, holographic-dictionary entries) that we leave for future work.

Phase 6c Wave 5 — RT/CH entropy bounds: classical vs W3 microscopic, and CH saturation across 2D CFTs

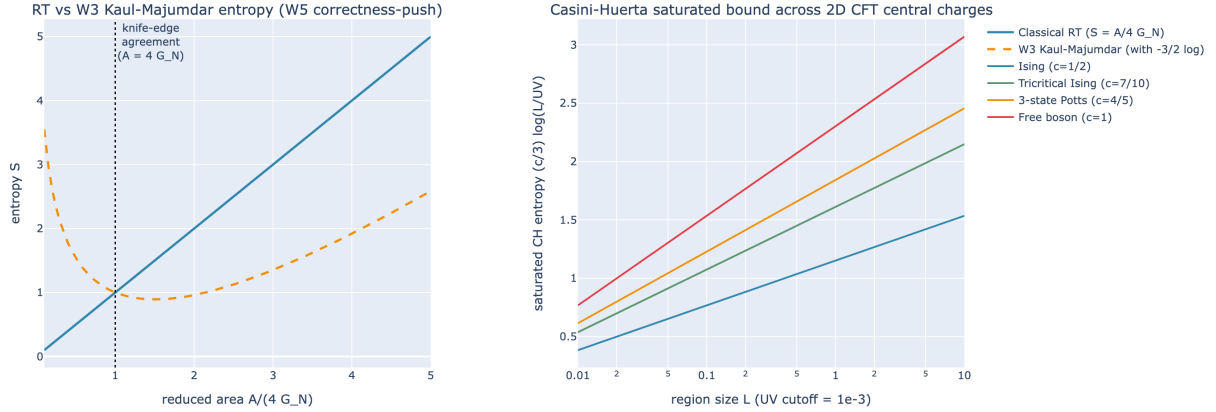


FIG. 1. *Ryu–Takayanagi area law versus Kaul–Majumdar microscopic entropy plus the saturated Casini–Huerta bound across central charges.* Left: classical Ryu–Takayanagi $S = A/(4G_N)$ (steel-blue, solid) versus Kaul–Majumdar $S = A/(4G_N) - (3/2)\log A/(4G_N)$ (amber, dashed) across reduced area $A/(4G_N) \in [0.1, 5]$. The knife-edge agreement at $A/(4G_N) = 1$ is highlighted; outside this point, the formulas necessarily disagree. Right: Casini–Huerta saturated bound $(c/3) \log(L/\epsilon)$ across central charges (2D Ising $c = 1/2$, tricritical Ising $c = 7/10$, 3-state Potts $c = 4/5$, free boson $c = 1$) at fixed $\epsilon = 10^{-3}$.

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